3D urban massing & shadow analysis

What this tool produces · *Visual and quantitative reports detailing exact shadow trajectories, solar exposure hours, and the visual obstruction caused by proposed building volumes on the surrounding public space.*

G2 · Proof of Fit

Dimensions

Spatial **Heritage** **Environmental**

aktan
With simplexity

Preparing the workshop · *3D urban massing & shadow analysis*

G2 · Proof of Fit

Why this tool matters

Defending the fundamental human right to light and sky; using geometry to hold developers accountable for the shadows they cast over the lives of their neighbors.

Evidence we produce

Visual and quantitative reports detailing exact shadow trajectories, solar exposure hours, and the visual obstruction caused by proposed building volumes on the surrounding public space.

Duration & Material

Duration · 1 to 3 weeks of modeling and simulation

Material · 3D modeling software (e.g., Rhinoceros, SketchUp), GIS analysis tools, cadastral maps

Reference case

Within the Kalasatama Digital Twins Project, the semantic 3D models were utilized to conduct precise "Sun Studies" and shadow analyses. Planners could instantly evaluate how the massing of new urban blocks would cast shadows on nearby residential areas, ensuring that proposed developments adhered to environmental comfort standards before granting planning permissions.

Suggested agenda · Brief (10 min) › Method walk-through (15 min) › Animation canvas (60–90 min) › Harvest (20 min)

Sun positions · shadows × times of year

Duration
1 to 3 weeks of modeling and simulation

Winter solstice

Dec 21 · 12h00 · long shadows

shadow render

Finding #1

Finding #2

Equinox

Mar 21 · 12h00 · medium shadows

shadow render

Finding #1

Finding #2

Summer solstice

Jun 21 · 12h00 · short shadows

shadow render

Finding #1

Finding #2

Hottest hour

Jul · 16h00 · west-facing heat

shadow render

Finding #1

Finding #2

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.